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ORIGINAL ARTICLE

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Divergent natural selection drives the evolution of reproductive isolation in an Australian wildflower

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Ecological speciation occurs when reproductive isolation evolves between populations adapting to contrasting environments. A key prediction of this process is that the fitness of hybrids between divergent populations should be reduced in each parental environment as a function of the proportion of local genes they carry, a process resulting in ecologically dependent reproductive isolation (RI). To test this prediction, we use reciprocal transplant experiments between adjacent populations of an Australian wildflower, *Senecio lautus*, at two locations to distinguish between ecologically dependent and intrinsic genetic reproductive barriers. These barriers can be distinguished by observing the relative fitness of reciprocal backcross hybrids, as they differ in the contribution of genes from either parent while controlling for any intrinsic fitness effects of hybridization. We show ecologically dependent fitness effects in establishment and survival of backcrosses in one transplant experiment, and growth performance in the second transplant experiment. These results suggest natural selection can create strong reproductive barriers that maintain differentiation between populations with the potential to interbreed, and implies a significant role for ecology in the evolutionary divergence of *S. lautus*.

KEY WORDS: **Ecological speciation, MCMCglmm, postzygotic isolation, reciprocal transplant, Senecio.**

Speciation is the process whereby populations become reproductively isolated from one another (Coyne and Orr 2004). Understanding the drivers of this process requires examination of the factors that prevent gene exchange between populations and their relative contributions to reproductive isolation (Sobel et al. 2010). Barriers to gene flow fall into two main categories: those that prevent mating between populations and therefore prevent the formation of hybrids (prezygotic), and those that prevent hybrids from mating if they are produced (postzygotic). These barriers may be independent of the environment in which they are measured (intrinsic) and arise from the evolution of genetic incompatibilities between populations (Orr and Turelli 2001), or act only in the local habitats of the diverging populations (extrinsic) and result from adaptation via divergent natural selection arising from ecological differences between populations (Schluter 2001). While intrinsic genetic isolating barriers may evolve in both ecological and nonecological modes of speciation, the presence of extrinsic postzygotic isolation, especially where intrinsic barriers are low, provides strong support for ecological speciation (Hatfield and Schluter 1999; Seehausen et al. 2014).

If extrinsic postzygotic isolation is present between diverging populations, it follows that intrinsic postzygotic barriers must be weak enough to produce viable hybrids in the laboratory, but ecological differences between populations create sufficient pressure on hybrids that they suffer fitness reductions under field conditions (Rundle and Whitlock 2001). This prediction arises because of two reasons: first, hybrid genotypes lack the adaptive phenotypes under selection in the habitat of each diverging population and therefore have reduced fitness relative to their parents (Nosil et al. 2005; Sobel et al. 2010); and second, reduction in hybrid fitness in parental but not in laboratory environments arises from adaptation to contrasting environments (Hatfield and Schluter 1999), and not from genetic-based incompatibilities that reduce fitness under stressful conditions (Rundle and Whitlock

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2001; Coyne and Orr 2004). Not surprisingly, separating hybrid fitness reductions arising solely from ecologically based divergent natural selection from those resulting from the evolution of intrinsic genetic incompatibilities between taxa remains a challenging exercise (Rice and Hostert 1993; Dettman et al. 2007; Egan and Funk 2009; McBride and Singer 2010). Moreover, early generation hybrids often show patterns of heterosis that may obscure the fitness effects of ecologically based selection (Lowry et al. 2008; Baack et al. 2016), further complicating the contributions of extrinsic postzygotic isolation to speciation.

In response to these difficulties, Rundle and Whitlock (2001) proposed an elegant approach to demonstrate the effects of ecological divergent natural selection on the evolution of reproductive isolation during speciation. They proposed that a comparison of fitness between reciprocal backcross genotypes across environments could separate ecological from intrinsic causes for extrinsic postzygotic isolation. This is because reciprocal backcross genotypes carry roughly the same number of crossbred loci, but differ in the proportion of locally adapted genes from each source population. As a consequence, reciprocal backcrosses are predicted to switch fitness ranks across environments, with the backcross populations having higher fitness in the conditions of the parent with which it is most closely related. Conceptually, this test states that strength of ecological extrinsic postzygotic isolation is a function of the proportion of locally adapted genes that individuals carry when exposed to that environment. F1 hybrids are expected to display intermediate fitness compared to parentals, but as mentioned before, heterosis can obscure this signal.

On the other hand, if intrinsic genetic incompatibilities are the predominant drivers of extrinsic reproductive isolation, there should be no switch in fitness rank, and F1 hybrids should show lower fitness than their parents. This is because fitness reductions arising from genetic incompatibilities should be independent of the specific environmental conditions to which hybrids are exposed (Lynch 1991; Rundle and Whitlock 2001; Rundle and Nosil 2005). Note that in experiments where only parental types and F1 hybrids are exposed to natural and laboratory conditions, distinguishing ecological from intrinsic causes of extrinsic postzygotic isolation is not possible (Coyne and Orr 2004). Although a few laboratory experiments have used Rundle and Whitlock's approach to examine the evolution of ecological extrinsic reproductive isolation in animals (Dettman et al. 2007; Egan and Funk 2009; Matute et al. 2009; McBride and Singer 2010) empirical examples in plants, and experiments using this method under natural conditions in the wild remain conspicuously rare (Rundle 2002).

Senecio lautus (Variable groundsel) is an outcrossing semiannual herbaceous plant native to Australia and the South Pacific islands. The species complex consists of multiple ecotypes where phenotypic differences between ecotypes correlate with environmental conditions (Ornduff 1964; Ali 1966; Radford et al. 2004). Coastal ecotypes of this species consist of multiple populations found in adjacent (parapatric) sand dunes (Dune ecotype) and rocky headlands (Headland ecotype), and several populations of an Island ecotype found on inshore islands. Headland and Dune pairs appear to have evolved repeatedly and independently on multiple occasions (Roda et al. 2013a,b), a pattern that is consistent with parallel adaptation to similar ecological conditions.

In the populations used in this experiment, estimates of genetic divergence are low (F_{st} = 0.08 for Coffs Harbour, and F_{st} = 0.09 for Lennox Head population; Roda et al. 2013a), which suggests recent divergence or gene flow between populations. However, a small fraction of the genome shows fixed genetic differences indicating that population boundaries are possibly maintained by divergent natural selection (Roda et al. 2013a). Previous work also discovered that local adaptation to coastal environments creates strong immigrant inviability between *S. lautus* ecotypes (Melo et al. 2014; Richards and Ortiz-Barrientos 2016). While ecotypes are genetically differentiated and occur in contrasting ecological conditions, intrinsic barriers to gene flow are weak, (Richards and Ortiz-Barrientos 2016), and populations still retain the ability to produce viable hybrid crosses when grown in a common environment (Ali 1966; Melo et al. 2014; Richards and Ortiz-Barrientos 2016). These characteristics provide an ideal opportunity to investigate the specific contributions of ecological divergent natural selection to the evolution of postzygotic reproductive isolation and thus to examine the role of ecological speciation in the origin of new plant species.

Here, we present results from two field-based reciprocal transplant experiments designed to test for the presence of ecological reproductive isolation between coastal populations of *S. lautus*. We include parental as well as F1 and F2 hybrids, and backcrosses between the Dune and Headland ecotypes at Lennox Head, and between Dune, Headland and Island ecotypes of *S. lautus* at Coffs Harbour. We ask whether ecological divergent natural selection can produce barriers to gene flow between these coastal ecotypes, as predicted by the ecological speciation hypothesis. We discuss how these results help us better understand how natural selection may drive speciation when there is opportunity for gene flow, and the similarities of the origin of species across the natural kingdoms.

Materials and Methods

FIELD LOCATIONS

All seeds used in these experiments were derived from seed collected in natural populations at Boambee beach (S 30° 18′ 45.28″, E 153° 8′ 21.43″, Dune type), Corambirra point (S 30° 18′ 44.09″, E 153° 8′ 41.51″, Headland type), and Mutton Bird Island (S 30° 18′ 19.67″, E 153° 8′ 57.27″, Isle type) at Coffs Harbour, NSW;

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and from the sand dunes (S 28° 47′ 1.23″, E 153° 35′ 38.56″, Dune type) and rocky headland (S 28° 48′ 47.22″, E 153° 36′ 19.15″, Headland type) at Lennox Head, NSW (see Table S1 for nomenclature distinguishing habitat vs. ecotype and population code). The natural populations at Coffs Harbour are monophyletic and occur in parapatry, with all seed from three populations collected within an 800 m radius. Plants from the Dune and Headland populations are separated by only a few metres. Seeds for Headland and Dune populations at Lennox Head were collected from sites separated by 3.4 km, and show greater genetic divergence, although the specific relationship depends on the loci used for phylogenetic reconstruction (Roda et al. 2013a; Melo et al. 2014).

SEED PRODUCTION

Seeds were collected from a minimum of 30 individuals per population with a minimum distance of 10 metres between plants to reduce the risk of collecting closely related individuals. Seeds were stored at four degrees in Tudor seed envelopes until required. Seeds were scarified and germinated on moist filter paper (whatman #4) in 4 cm petri dishes. Petri dishes were kept in the dark for two days before being transferred to a controlled temperature room where light was maintained at a 50:50 day-night cycle at 25°C. After a week, germinated seeds were transferred to the glasshouses where they were transplanted into 85 mm square pots. Controlled crosses were performed in the glasshouses at The University of Queensland between January 2012 and November 2013 by rubbing flower heads from parental plants together to transfer pollen; we repeated this procedure for three days for each cross so most florets contributed seed to subsequent generations. All seeds used in the field experiments were from full-sib families created by crosses in the glasshouse. Families were generated by crosses within ancestral populations (Parentals, P), between populations (F1 hybrids, F1), within F1s (F2 hybrids, F2), and between F1s and Parentals (Backcrosses, BC). These crosses are referred to from here on as genotypes, with the abbreviated names above followed by a population code to indicate the cytoplasmic origin (see Fig. 1, Table S1 for full explanation of notation). Few Backcross families were created between Dune-Isle F1s and Island parents (BCID) due to limited flowering time overlap in the glasshouse, so we excluded this Genotype from the experiment. The reciprocal backcross between Dune-Isle F1s and Dune parents (BCDI) was included in the experiments. Crossing design and the populations included in these experiments are outlined in Figure 1.

FIELD TRANSPLANT

The transplant was started on the 20th April 2014 in the Coffs Harbour environments, and on the 18th of March 2014 in the Lennox head environments. Both transplants occurred within the locations of the initial seed collections. Four plots were established in each of the three Coffs Harbour environments, each treated as an individual block. Twelve plots were established in each of the two Lennox Head environments, with two contiguous subplots contributing to a block. Vegetation and litter were removed from the plots before planting. Seeds from each full-sib family were organised into a completely randomised block design before transfer to the field sites. Seeds were glued to toothpicks using Selleys Quick fix supa glue gel and organised into polystyrene grids in the layout of each field block (Melo et al. 2014). One seed from each family was planted into the corresponding cells of 400 × 600 mm grids with the seed placed ~2 mm below the soil surface. In the Coffs Harbour transplant, we included four individuals each from 15 families in each environment (12 individuals per family in total), leading to 60 individuals per genotype per environment. As our design included all possible crosses between the three populations, we planted 14 different genotypes (Fig. 1), leading to 1020 seeds per environment and 3060 seeds across the whole transplant. In Lennox Head we had six genotypes, and we planted six seeds per family from 32 backcross families in each direction (BC-H, BC-D), 30 F1 families, 30 parental families 27 F2 families per environment. This resulted 1086 seeds planted in each environment, for a total of 2172 seeds across the whole transplant. In total, we planted 5232 seeds across the two experiments.

All plots were watered and covered with shade cloth in the early stages. These conditions mimicked natural germination conditions at these localities. Each block was watered daily with 1 liter of water per block (equivalent to ~4 mm of rainfall) for the first 21 days at both locations. At Coffs Harbour, plots were covered with 50% shade cloth for the first four weeks, after which it was replaced with open weave bird netting. In the sand dunes at Lennox Head, shade cloth was replaced after 100 days. The Lennox Head experiment ran for 11 months with germinations and deaths recorded monthly. The Coffs Harbour experiment ran for eight months with fitness recorded daily for the first 21 days of the experiment, after which fitness was recorded weekly until harvest on day 161 at the Island site, and day 198 at the Dune and Headland sites. Fitness measures recorded were germination, mortality, establishment, and the presence of a buds and flowers. Establishment was taken as the production of seven leaves at Coffs Harbour and 10 leaves at Lennox Head as a measure of a plant's ability to transition between early and late life-history stages. Glasshouse experiments have shown that the production of 7–10 leaves is a good indicator of establishment and the transition from juvenile to adult in *S. lautus* (Bernal-Franco 2015). The experiments were ceased when plants began to senesce. Plants from the Coffs Harbour experiment were clipped at ground level and individually sealed in ziploc bags and transported back to the University of Queensland where we recorded harvest mass as a measure of growth performance.

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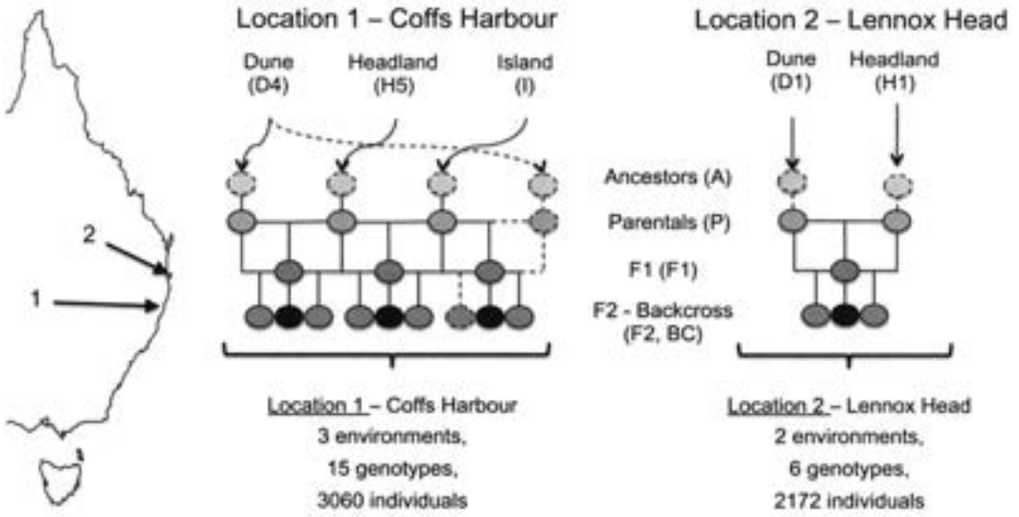


Figure 1. Location and crossing design used in transplant experiments.

Table 1. Anova of Harvest mass of Backcross genotypes organized by population comparisons.

Populations	Predictors	Df	Sum Sq	Mean Sq	*F* value	Pr(>*F*)
D-I	Population	1	1.572	1.572	1.959	0.176
	Environment	1	3.543	3.543	4.415	0.048
	Block	1	0.156	0.156	0.195	0.664
	pop:env	1	0.288	0.288	0.358	0.556
	Residuals	21	16.851	0.802		
H-I	Population	1	0.808	0.808	1.854	0.185
	Environment	1	1.813	1.813	4.161	0.051
	Block	1	0.136	0.136	0.312	0.581
	pop:env	**1**	**6.207**	**6.207**	**14.241**	**0.001**
	Residuals	27	11.767	0.436		
H-D	Population	1	0.049	0.04965	9.344	0.055
	Environment	1	0.00004	0.00004	0.008	0.935
	Block	1	0.161	0.16078	30.257	0.012
	pop:env	**1**	**0.058**	**0.05867**	**11.04**	**0.045**
	Residuals	3	0.016	0.00531		

Significant interaction terms demonstrate ecologically dependent fitness and are highlighted in bold.

STATISTICAL ANALYSIS

To determine fitness differences between the parental and hybrid populations, we defined three components of fitness: establishment, survival, and growth performance. While related, these three different measures of fitness allow distinctions to be made across multiple life-history stages, and therefore elucidate whether selection acts in a uniform way across the lifespan of a plant, or whether there are particular developmental stages at which ecological factors have greatest effects on fitness. While flowering and seed production is the best estimate of lifetime fitness, we were unable to get reliable estimates of these traits due to low sample size of flowering plants across the experiment. As mentioned, no plants flowered in the Dune environment. Across the two environments where we have data on flowering, there was a very strong correlation between growth and flower count ($r = 0.78$). We therefore used harvest mass in these analyses, as this measure increases sample sizes and provides a reasonable proxy of flower production. We outline the methods for the three fitness components in chronological order from seedling emergence to harvest.

Probability of establishment

To calculate the fitness of cross types in the field we used a conditional generalized linear-mixed model in a Bayesian framework to estimate the mean probability of germination and establishment for each genotype. These models estimate the probability

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of an event based on known conditions that may be related to the event. In this case, we estimate the probability of each genotype passing through the initial establishment life-history stage using information about germination. This process takes into account that missing values of later life-history fitness components, such as leaf production, are not missing at random, but may result from systematic patterns in early life-history stages such as germination (Steinsland et al. 2014).

We modeled fitness over life-history stages using the conditional response variable (Emergence –> Survival to day 100 –> Ability to produce 10 leaves) for Lennox Head, and (Germination –> Ability to produce seven leaves) for Coffs Harbour, where the arrow (–>) represents the direction of conditionality between each transition. The differences in the models between environments reflect different logistical issues between the two experiments (e.g., different frequency of data monitoring among transplant sites), but the measures of seven and 10 leaves are equivalent of life-history transition in each population. Analyses were conducted using the R package MCMCglmm (Hadfield 2010). We analyzed two models: one to test for local adaptation and one to test for ecologically dependent backcross fitness. To test for local adaptation we used the conditional response variables outlined above and included environment and the percentage of local genes for each cross type as fixed effects and family (full-sibling family) and block as random effects. To test for ecologically dependent fitness effects we used cross type (ancestor/backcross) and environment as fixed effects, and family and block as random effects.

We implemented Markov Chain Monte Carlo sampling to estimate the posterior probability distribution for the model parameters. The number of iterations and the thinning interval were increased until the effective sample size exceeded 85% of the desired 1000 mark for all parameters estimated. We used a Cauchy prior distribution (Gelman et al. 2008), incorporating the mean of each conditional level, and repeated the analyses while adjusting the variance to excessively large and small values output to ensure it did not change the model output and that we had confidence of an uninformative prior. Our final models included one million iterations, with a burn-in of one hundred thousand and a thinning interval of one thousand to remove autocorrelation between MCMC samples. To verify model performance we compared the distribution of the observed data to the posterior predictive distribution. All models indicated alignment between the means and variance of the two distributions validating model reliability.

Survival analysis

Probability of survival was modeled using a maximum likelihood estimation of parametric survival using the coxme package (Therneau 2012) in R (version 3.0.3). In the Coffs Harbour experiment, the survival function (response variable) was created using the day of germination, day of death and whether the individual was dead or alive at the censor point, day 161. Population was included as a fixed effect with block and family included as random effects. In the Lennox Head experiment, we analyzed time to death as the response variable censored by survival at day 320. Ecotype was included as a fixed effect and block and family were included as random effects. Separate analyses were carried out for each environment as the comparison of interest is between genotypes within a single environment.

Performance of survivors

We analyzed growth in survivors from the Coffs Harbour transplant to determine whether there was variation in the relative fitness of individuals that survived to the end of the experiment. We recorded harvest mass and dry weight to determine biomass accumulation. Very low biomass for many individual plants meant that we could not reliably account for measurement error from our dry mass measurements, so only harvest mass is analyzed here. To determine difference in growth between backcrosses and F2 hybrids against the Parental populations we fit a linear-mixed effects model using restricted maximum likelihood (REML) using the lmeTest package (Kuznetsova et al. 2013) in R. We used harvest mass as the response variable, which was log+1 transformed to ensure a normal distribution, and included population as a fixed effect with block and family as random effects. Family was removed from the model when testing Dune and Island populations in the dune environment, due to singularity problems arising from differences between the families surviving in each environment. Tests were done on backcrosses between two populations within one environment. To determine whether there was an environment × genotype interaction in backcross growth, we used ANOVA with log+1 transformed harvest mass as the response, and backcross population, environment, population × environment interaction and block as predictors. These analyses were performed between reciprocal backcrosses in their parental environments.

Overall, we infer patterns of ecologically dependent isolation as intrinsic when the rank of high to low fitness does not vary between environments, and as ecological reproductive isolation when we observe a reversal in the fitness rank between reciprocal backcross types in alternative environments.

Results

PROBABILITY OF ESTABLISHMENT

In general, establishment was lower in all Coffs Harbor sites than at Lennox Head. Seeds (seedlings) with 100% local genes had higher establishment success than those with 0% local genes in all experimental locations (Fig. 2). In the two sand dune sites, F2

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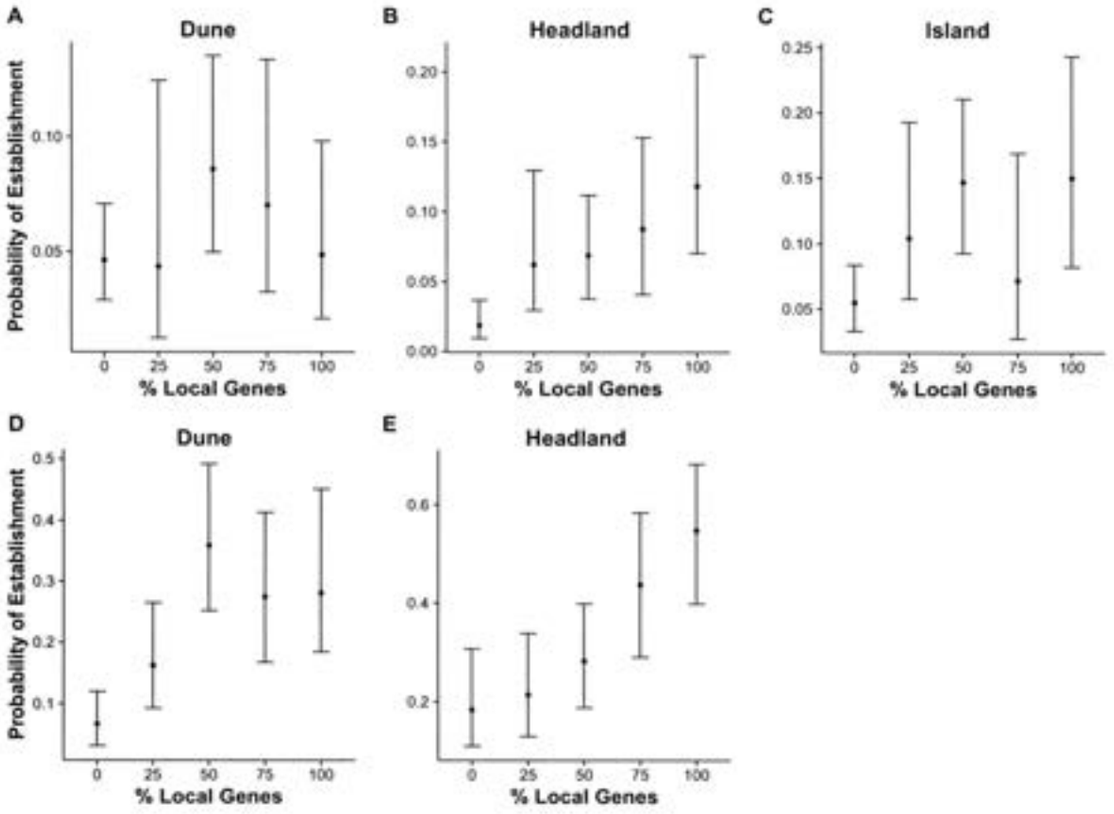


Figure 2. Posterior probability of establishment based on percentage of local genes. Coffs Harbour included 17 genotypes, Lennox Head included six genotypes.

Bayesian estimates of the probability of establishment based on conditional generalized linear-mixed models. Points show the mode of the posterior distribution while bars show the 95% credible intervals. Establishment is explained by the percentage of local genes in transplant locations at Coffs Harbour (A, B, C) and Lennox Head (D, E). Local genotypes consistently have high probability of establishment in all but the Coffs Harbour Dune (A). The pattern is strongest in Headland environment (B, E), while hybrids with 50% local genes have a high probability of establishment in both Dune and Island environments (A, C, D).

hybrids possessing 50% dune genes had the highest probability of germination and establishment (Fig. 2). In the rocky headland sites, the probability of establishment increased in a roughly additive manner with an increasing proportion of local genes, however this pattern was not observed in the sand dunes or the island sites. In the island site, F2 hybrids with 50% local genes had higher probability of establishment than backcross individuals with 75% local genes (Fig. 2).

In the Lennox Head sites, the local parental population had the highest probability of establishment, while the immigrant parental had the lowest. Backcross fitness displayed the predicted switch of rank under the ecological speciation hypothesis, with the backcross to the local population having a higher probability of establishment in each case.

In all four pairwise comparisons, patterns of local adaptation were present between parental genotypes (Fig. 3), although there were contrasting responses in the backcross genotypes. The backcross to Headland between Headland and Isle individuals performed better than the corresponding backcross to Isle individuals in both environments, suggesting that differences in establishment between these two backcross genotypes are intrinsic rather than ecologically dependent.

As the BCID backcross genotype was excluded from the experiment, we contrasted backcross and F2 fitness between Dune and Island populations, as they are both recombinant hybrid genotypes that differ in the proportion of genes from each parental population. As we expected, F2 fitness did not change between environments while the backcross to Dune showed higher fitness in the sand dunes, and lower fitness in the Island site, however results in the Dune site were ambiguous, due to low establishment across all genotypes.

We also compared F1 and F2 hybrids with the parents in each environment to assess the presence of heterosis in establishment (Fig. S2). A pattern of heterosis should decrease the effects

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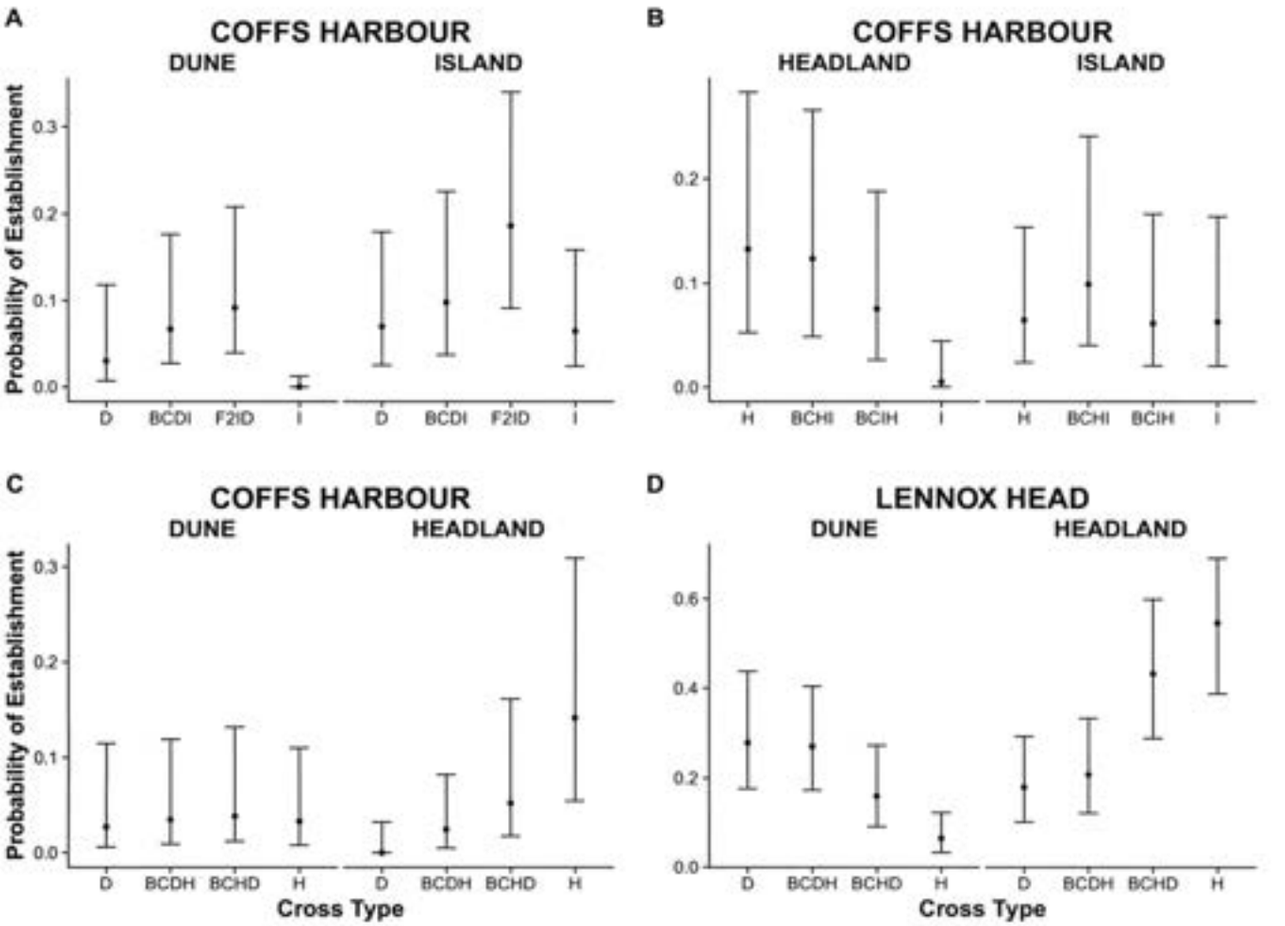


Figure 3. Extrinsic and intrinsic fitness effects on probability of establishment between ancestral ecotypes and reciprocal backcrosses. Bayesian estimates of the probability of establishment based on conditional generalized linear-mixed models. Ecologically dependent fitness is seen between H-D and H-I comparisons in the Headland environment but not the alternative environments (panels B, C), and in D-H comparisons at Lennox Head (D). D-I comparisons reveal a pattern of heterosis. (A). Note the use of F2 hybrids instead of BCMB in this comparison (see methods and Fig. 1).

of extrinsic postzygotic isolation as outcrossing produces a fitness advantage to hybrid genotypes, and also tests for the effect of intrinsic genetic incompatibilities. In general, the probability of hybrid establishment was higher than both local and immigrant parents in both the Dune and Island environments, while in the headland environments, immigrant parentals had the lowest probability of establishment, local headland parents the highest establishment rates and F1 and F2 hybrids at intermediate levels (Fig. S2).

SURVIVAL

Survival was calculated as the number of days between germination and death censored at day 161 of the experiment in Coffs Harbour. There was little to no effect of genotype on survival across the three sites (Table S2). In the sand dunes, survival was low with no significant differences between genotypes, although the random effect of block explained 66% of variance in individual survival (Table S3). This is not surprising, because two of the four blocks died out completely and suggests environmental heterogeneity may have had a substantial effect on survival within the Dune habitat. In the rocky headlands, BCHD and F1DH genotypes had lower risk of dying compared to the ancestral headland genotype (relative risk = 0.46, $z = -2.14$, $p = 0.032$, and 0.49, $z = -2.02$, $P = 0.04$). In contrast, BCIH individuals had the lowest risk of perishing (relative risk = 0.23, $z = -3.52$, $p = 0.0004$, Table S2).

Random effects accounted for less variance in survival in the rocky headland than in the sand dunes, although both random factors are influential with experimental block explaining 26% or variance in survival, and Family explaining 25%. In the Island site, only the F2HI genotype had significantly lower survival risk than ancestral Island individuals (relative risk = 0.36, $z = -2.09$,

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$P = 0.037$); however the variation in survival attributed to the random factors family (46%), and block (37%), suggests a substantial effect of within population variation and microhabitat on individual survival at the Island site.

In contrast to the three populations from Coffs Harbour, genotype had a significant effect on survival at Lennox Head sites. The local parental population from each habitat had the lowest risk of death, while the immigrant parent had over two times higher risk of dying in that environment (Table S2). Furthermore, in both environments, risk increased with the proportion of foreign genes in the genotype, except for the F1DH genotype in the Dune habitat, which was not significantly different to the Dune parent (relative risk = 0.88, $z = -0.77$, $P = 0.44$, Table S2).

PERFORMANCE OF SURVIVORS

Comparisons between Headland and Island, and Dune and Headland populations showed a pattern of ecologically dependent fitness in backcross genotypes (Fig. 4). The significant backcross type-environment interaction term between Headland-Isle ($F_{1,27} = 14.24$, $P = 0.001$) and Headland-Dune ($F_{1,3} = 11.04$, $P = 0.045$) showed a switch in the fitness rank of reciprocal backcrosses between environments. There was no significant interaction effect between backcrosses between Dune and Island genotypes ($F_{1,21} = 0.36$, $P = 0.556$) (Table 1).

Mixed effects regression models of between-population crosses within a single environment only revealed a significant population effect in the comparison of Headland and Isle crosses in the island ($F = 5.87_{4,\ 15.98}$, $P = 0.004$). There was a marginally significant effect of source population between Dune and Headland crosses in the sand dunes ($F = 4.42_{\ 4,5.99}$, $P = 0.055$), but as less than 10 individuals from each genotype survived to the conclusion of the experiment, we interpret this result cautiously.

There was a strong pattern of immigrant inviability between Dune-Headland and Headland–isle comparisons. In each case, immigrant parentals had the lowest harvest mass, and also the lowest sample size meaning fewer individuals survived to maturity (Fig. 4). The local parental populations had the highest harvest mass in all cases, showing beneficial effects of local adaptation on performance.

Discussion

Ecological speciation describes the process in which populations become reproductively isolated by divergent natural selection (Hatfield and Schluter 1999; Rundle and Nosil 2005). The presence of ecologically dependent postzygotic isolation is therefore consistent with the predictions of speciation resulting from divergent natural selection. This occurs when hybrid genotypes display phenotypes that are intermediate between the optimum phenotype of either parent and as such display lower fitness in the two parental habitats (Hatfield and Schluter 1999). Reciprocal backcross genotypes are fundamental to isolating environmentally dependent fitness effects from those arising from intrinsic genetic incompatibilities: backcross genotypes are more genetically similar to one of the parents, and therefore should have relatively high fitness in the environment of that parent they are more similar to (Rundle and Whitlock 2001).

While there are a number of studies that support a link between local adaptation to reproductive isolation (Nagy and Rice 1997; Hall and Willis 2006; Lowry et al. 2008), few studies have measured the relative fitness of backcross genotypes in the wild, a result that is crucial to isolating the effects of divergent natural selection on populations with the potential to interbreed (Rundle and Whitlock 2001). Our work here uses a method to make distinction between ecologically dependent and nonecologically dependent RI, which builds on a growing body of evidence from insects (Egan and Funk 2009; Kuwajima et al. 2010; McBride and Singer 2010) and fish (Rundle 2002; Fuller 2008) to show that ecological reproductive isolation is a potentially important driver in the divergence of plant populations in the wild.

Ecological factors have been implicated in previous work on the origins of ecotypes in the *S. lautus* species complex (Roda et al. 2013b; Melo et al. 2014). The repeated parallel divergence of Dune and Headland ecotypes in the system is a pattern best described by the action of natural selection driving adaptation to these two environment types (Roda et al. 2013a). In Lennox Head, the signature of ecologically dependent reproductive isolation was present across two components of fitness, establishment, and survival. There was a strong signature of local adaptation in parental genotypes and backcross fitness was higher when the backcross was in the direction of the local genotype. Here, it seems that ecological factors both prevented establishment of immigrant and hybrid genotypes, and increased the risk of mortality of any immigrants that do establish, leading to ecologically dependent reproductive isolation.

In the Coffs Harbour sites, establishment was ecologically dependent in the Headland site, while we saw a pattern of hybrid vigor in the Dune and Island environments. We also observed no significant differences in survival between genotypes in any of the three environments when genotype was included as a main effect in survival models, although the signature of ecologically dependent postzygotic isolation is most clearly seen in the measures of growth in plants that both established and survived. Despite a small sample size, the significant interaction between population and environment shows the diagnostic switching of fitness rank between backcross types depending on the proportion of local genes. This result suggests that ecological differences between the three environments do not prevent establishment or survival of immigrants but create barriers to gene flow by reducing growth and performance in immigrant and hybrid genotypes. This may be

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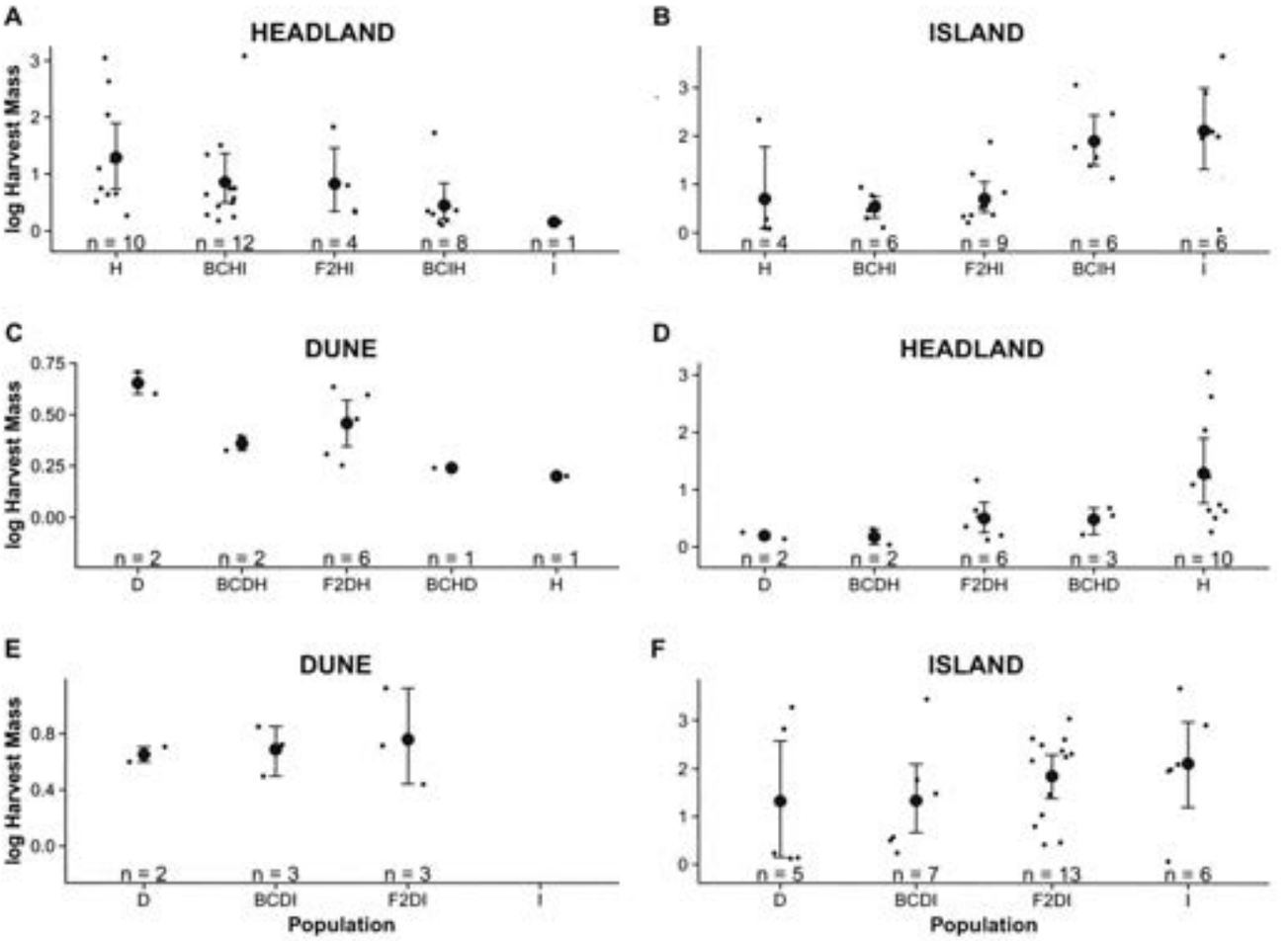


Figure 4. Comparison of growth performance between recombinant hybrids and parental populations from Coffs Harbour. Harvest mass is log(n + 1) transformed and describes the total biomass accumulation over the lifespan of all surviving plants. Panels A and B show the reciprocal transplant between Headland and Island derived populations in each Headland and Island sites. Panels C and D describe the Dune-Headland comparison and Panels E and F describe Dune-Island comparison, respectively. Dotted lines show significant interactions between backcross type and environment, meaning the mass accumulation was highest in the backcross to the local population that is a diagnostic pattern associated with ecologically dependent reproductive isolation.

due to the effects of heterosis masking the signature of ecological selection in early life-history stages, but also suggests that the fitness benefits of heterosis are removed by the stage of maturity. However, the small sample sizes in measures of biomass accumulations are a result of very low survival to maturity across all populations. This is particularly evident in the sand dunes, where less than 30 individuals (of 1020 planted seeds) survived to the censor point of the experiment. Low survivorship to maturity suggests that all three environments are harsh and that any signature of divergent selection on establishment and survival is obscured by general harsh conditions that affect all genotypes (as discussed by Blanquart et al. 2013). Over the duration of the experiment, Coffs Harbour received around a third of its mean monthly rainfall, which meant conditions on the more exposed areas of the Dune environment were particularly harsh. We acknowledge that there are limitations in the conclusions that can be drawn from such limited sample sizes; however, the observed trends are consistent across multiple analyses suggesting that the general trends are not random.

One explanation for the observed differences between experiment locations may be due to the increased genetic divergence between Lennox Head populations compared with the divergence between Coffs Harbour populations. Based on neutral SNPs (Roda et al. 2013a), Dune, Headland, and Island ecotypes at Coffs Harbour form a monophyletic group within the phylogeny of *S. lautus* in Australia, while the Lennox Head ecotypes are more genetically diverged. This is consistent with evidence from insects (Presgraves and Noor 2002) and plants (Moyle et al. 2004; Scopece et al. 2008), where the initial barriers to gene flow between diverging populations arise due to extrinsic postzygotic

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isolation, and the strength of postzygotic isolation increases with genetic distance between taxa (Coyne and Orr 1989). Observation of extrinsic barriers during early points on the speciation continuum and the absence of clear intrinsic barriers, as outlined here in *Senecio*, is also a strong indicator of speciation driven by divergent natural selection. This model contrasts with those where speciation is driven by intrinsic factors and extrinsic barriers arise at later stages along the speciation continuum (Seehausen et al. 2014).

While ecologically dependent postzygotic isolation is a fundamental prediction of ecological speciation theory, few empirical examples exist where the specific barriers are isolated from other forms of reproductive isolation. We show in this work that multiple fitness measures across life-history stages show patterns of ecologically dependent postzygotic isolation. Further, these patterns are consistent between replicate ecologically divergent parapatric populations, and the strength of these barriers increases with the degree of evolutionary divergence between populations. Whether this form of ecologically based reproductive isolation will facilitate the evolution of other barriers in the future remains an open question. Our work suggests that the underlying mechanisms of ecological speciation may be contribute to evolutionary divergence within the *Senecio lautus* species complex and provide a robust example of how ecological differences between populations can lead to reproductive isolation.

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T.J.R. and G.M.W. collected seeds from wild populations, performed experiments, and analyzed data. T.J.R., G.M.W., K.M., and D.O.: designed research. T.J.R. and D.O. wrote the article with input from all authors.

Authors declare no conflict of interest.

DATA ARCHIVING

The doi for our data is 10.5061/dryad.17h0t.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Table S1. Details of populations included in Field transplant experiment.
Table S2. Table of fixed effects from Cox mixed effects model.
Table S3. Table of Random effects from Cox mixed effects model.
Figure S1. Patterns of Heterosis between parents and F1 and F2 Hybrids.

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